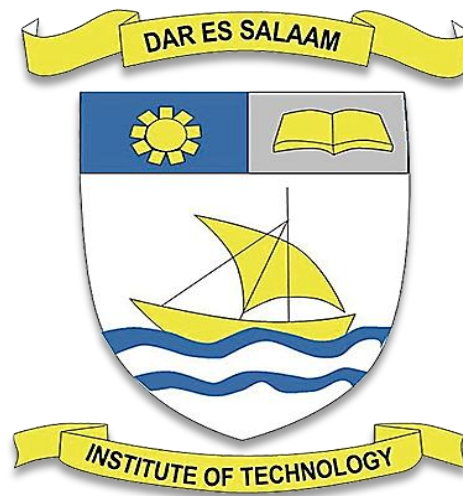


DAR ES SAALAM INSTITUTE OF TECHNOLOGY



**COMPUTER ENGINEERING DEPARTMENT
BACHELOR DEGREE IN COMPUTER ENGINEERING
NTA LEVEL 8**

**PROJECT TITLE: DEVELOPMENT OF SMART MARKET AND
PRICE DECISION SUPPORT SYSTEM: CASE OF
ENTREPRENEURS, FARMERS and BUYERS AT
Using Multi-Source Market Data Analytics and USSD Technology**

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MARCH 1, 2026

DECLARATION

We, Nawal Ismail and Stephen Nziku, hereby declare that the project report entitled **Development of Smart Market and Price Decision Support System in case of Entrepreneurs, Farmers and Buyers using Multi-Source Market Data Analytics and USSD Technology** is the result of our own original work carried out under the supervision of **Mr. Alfred Kajirunga**. This report has been prepared in partial fulfillment of the requirements for the award of the relevant academic qualification.

We further declare that this project has not been submitted either in whole or in part to any other university, college or institution for the award of any degree, diploma, certificate or any other academic qualification. The ideas, findings, analyses, designs and conclusions presented in this report are the outcome of our own efforts except where explicit reference has been made to the work of other authors and researchers.

In conducting this project, we adhered to the principles of academic integrity, honesty and ethical research practices. All sources of information, including books, journals, articles, reports, websites and other materials consulted during the development of this project have been appropriately acknowledged and cited in accordance with accepted academic standards. Any contribution made by other individuals or organizations has also been duly recognized.

We take full responsibility for the content of this report including any errors or omissions that may be found therein. We affirm that the work presented reflects our knowledge, understanding and practical application of information and communication technology concepts in addressing real-world challenges related to market information accessibility, price analysis and decision support for entrepreneurs, farmers and buyers through the integration of multi-source market data analytics and USSD technology.

Therefore, we solemnly declare that this report is our authentic work and is submitted for academic examination and evaluation.

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ABSTRACT

Access to accurate, timely and reliable market information remains a significant challenge for entrepreneurs, farmers, buyers and other business stakeholders, particularly in agricultural communities such as Ifakara. Many market participants rely on fragmented information sources, manual price collection methods and informal communication channels, resulting in poor decision-making, price uncertainty, reduced profitability and missed business opportunities. Furthermore, limited internet accessibility in rural areas restricts many farmers from accessing real-time market information that could support effective production and marketing decisions.

This project presents the development of a Smart Market and Price Decision Support System using Multi-Source Market Data Analytics and USSD Technology, with Ifakara serving as the case study area. The system is designed to collect, analyze, predict and disseminate market information from multiple sources to support entrepreneurs, farmers, buyers and other stakeholders in making informed and data-driven decisions. The proposed system integrates market data from traders, market officers, online marketplaces, social media platforms, e-commerce platforms and publicly available APIs. The collected information is processed and stored in a centralized database to support market analysis, price prediction and decision-making activities.

Machine learning techniques are applied to identify market trends, forecast demand patterns and predict future commodity prices. The system generates intelligent recommendations that assist users in production planning, pricing strategies, purchasing decisions and identification of business opportunities. To improve accessibility among rural users, the system incorporates a USSD-based interface that enables farmers and other stakeholders without internet access to retrieve market prices, forecasts and recommendations using ordinary mobile phones.

The system further provides dashboards, reports, visualizations and analytical tools that facilitate market monitoring and strategic decision-making. The implementation of the proposed system is expected to improve market transparency, reduce information asymmetry, enhance business competitiveness and promote data-driven decision-making among entrepreneurs, farmers and buyers in Ifakara and similar agricultural communities. Ultimately, the system contributes to improving market efficiency and supporting sustainable economic development through the application of modern information technologies and artificial intelligence techniques.

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TABLE OF CONTENTS

DECLARATION	ii
ABSTRACT	iv
ACKNOWLEDGEMENT	v
LIST OF FIGURES	viii
LIST OF TABLES.....	ix
LIST OF FIGURES	x
LIST OF ABBREVIATIONS.....	x
CHAPTER ONE	1
INTRODUCTION	1
1.1. BACKGROUND INFORMATION	1
1.2. PROBLEM STATEMENT.....	4
1.3. PROJECT OBJECTIVE	5
1.3.1. GENERAL OBJECTIVE	5
1.3.2. SPECIFIC OBJECTIVE	5
1.4. SCOPE OF THE STUDY	6
1.5. SIGNIFICANCE OF THE STUDY	6
1.6. LIMITATION OF THE STUDY	7
1.7. CHAPTER SUMMARY	7
CHAPTER TWO	9
LITREATURE REVIEW	9
2.1. INTRODUCTION	9
2.2. EXISTING MARKET INFORMATION SYSTEMS	10
2.2.1. LITREATURE REVIEW GAP IN EXISTING MARKET INFORMATION SYSTEMS	11
2.2.2. BLOCK DIAGRAM OF THE EXISTING SYSTEM (LEOLEO APP)	12
2.3. PROPOSED SYSTEM OVERVIEW	13
2.3.1. KEY COMPONENTS OF THE PROPOSED SYSTEM	14
2.3.2. BLOCK DIAGRAM OF THE PROPOSED SYSTEM	16
CHAPTER THREE	18
METHODOLOGY	18
3.1. INTRODUCTION	18
3.2. SYSTEM DEVELOPMENT	18

3.2.2. JUSTIFICATION FOR CHOOSING AGILE METHODOLOGY	22
3.2.4. BENEFITS OF AGILE FOR THIS PROJECT	22
3.2.3 AGILE IMPLEMENTATION STEPS	23
3.3. DATA COLLECTION	24
REFERENCE	25

LIST OF FIGURES

Figure 2.1 Block Diagram of M-Klilimo system	13
Figure 2.2 Block diagram of a proposed system.....	17
Figure 3.1 Block diagram of a proposed system.....	19

LIST OF TABLES

Table 1 Existing Market information system.....	10
Table 2 Features and Purpose of Each Sprint in Agile Methodology.....	23

LIST OF FIGURES

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
DIT	Dar es Salaam Institute of Technology
FAO	Food and Agriculture Organization
GDP	Gross Domestic Product
MIS	Market Information System
SMS	Short Message Service
TMX	Tanzania Mercantile Exchange
USSD	Unstructured Supplementary Service Data

CHAPTER ONE

INTRODUCTION

1.1. BACKGROUND INFORMATION

The advancement of Information and Communication Technology (ICT) has significantly transformed the way information is collected, processed, analyzed, and disseminated across various sectors of the economy. Modern technologies such as artificial intelligence (AI), machine learning (ML), big data analytics, cloud computing, and mobile communication technologies have enhanced the ability of organizations and individuals to make informed decisions based on accurate and timely information (United Nations Conference on Trade and Development (UNCTAD), 2024). In today's data-driven economy, access to reliable market information has become a critical factor in improving productivity, competitiveness, profitability, and sustainable economic development.

Agriculture remains one of the most important sectors in developing countries, particularly in Sub-Saharan Africa, where it serves as a primary source of income, employment, and food security for millions of people (Food and Agriculture Organization, 2022). In Tanzania, agriculture contributes significantly to the national economy and supports the livelihoods of a large proportion of the population. Farmers, traders, entrepreneurs, and buyers depend heavily on market information to make decisions related to production planning, pricing, marketing, procurement, and investment. However, despite the importance of market information, many stakeholders continue to face challenges in accessing accurate, timely, and reliable information regarding commodity prices, market demand, supply conditions, consumer preferences, and emerging business opportunities (Galtier et al., 2014).

Market information plays a crucial role in reducing information asymmetry between buyers and sellers. Information asymmetry occurs when one party possesses more information than another during market transactions, resulting in unfair pricing, poor bargaining power, and inefficient market operations. In many rural communities, farmers often sell their produce without knowing prevailing market prices in different locations. As a result, middlemen and traders may take advantage of this information gap by offering lower prices, which ultimately reduces farmers' income and profitability. Similarly, buyers and entrepreneurs may face difficulties in identifying suitable suppliers, understanding market trends, and making strategic purchasing decisions.

The emergence of digital technologies has provided new opportunities for addressing these challenges through the development of Market Information Systems (MIS). A Market Information System is a computerized platform designed to collect, process, store, analyze, and disseminate market-related information to support decision-making among stakeholders (Galtier et al., 2014). Such systems can aggregate data from multiple sources including markets, traders, government agencies, social media platforms, e-commerce websites, online marketplaces, and public databases. By providing real-time access to market information, these systems contribute to improved market transparency, efficient resource allocation, and better economic outcomes.

In recent years, the increasing availability of digital data has encouraged the adoption of data analytics and machine learning techniques for market analysis and forecasting. Machine learning is a branch of artificial intelligence that enables computer systems to learn patterns from historical data and make predictions without being explicitly programmed for every task (Han et al., 2023). Through machine learning algorithms, large volumes of market data can be analyzed to identify hidden trends, predict future commodity prices, estimate demand fluctuations, detect consumer behavior patterns, and generate intelligent recommendations. These capabilities have made machine learning an essential tool in modern decision support systems used in agriculture, commerce, finance, and supply chain management (Vijay Kotu & Bala Deshpande, 2019).

Price prediction has become one of the most important applications of machine learning in market intelligence systems. Accurate price forecasts help farmers determine the best time to sell their products, assist buyers in planning purchases, and enable entrepreneurs to identify profitable business opportunities (Food and Agriculture Organization (FAO), 2023). By analyzing historical market prices together with factors such as seasonal patterns, demand trends, weather conditions, and economic indicators, machine learning models can generate forecasts that support informed decision-making and reduce uncertainty in market transactions.

Despite the growth of internet-based services, internet accessibility remains limited in many rural areas of Tanzania. A considerable number of farmers and rural stakeholders still rely on basic mobile phones that do not support advanced internet applications (Tanzania Communications Regulatory Authority [TCRA], 2024). Consequently, web-based systems alone may not adequately serve all intended users. USSD (Unstructured Supplementary Service Data) technology offers an effective alternative by enabling users to access information and services through mobile phones without requiring internet

connectivity. USSD services are widely available, cost-effective, and accessible on both smartphones and feature phones, making them particularly suitable for rural communities (Food and Agriculture Organization (FAO), 2023).

Ifakara, located in the Morogoro Region of Tanzania, is an area where agriculture and small-scale business activities contribute significantly to local economic development. Farmers, entrepreneurs, traders, and buyers operating within Ifakara frequently require accurate market information to make informed decisions regarding production, pricing, marketing, and investment activities. However, market information is often obtained through informal communication channels such as personal networks, local traders, physical market visits, and word-of-mouth exchanges. These methods may not always provide timely, reliable, or comprehensive information, leading to poor decision-making and reduced economic opportunities (Ministry of Agriculture Tanzania, 2024).

Furthermore, existing market information platforms often focus on isolated sources of data and may not provide integrated market intelligence capable of supporting comprehensive decision-making. The lack of a centralized platform that combines market data collection, data analytics, price forecasting, recommendation generation, and USSD-based accessibility limits the ability of stakeholders to fully benefit from available market information. This challenge highlights the need for a smart and integrated market information solution that leverages modern technologies to improve market transparency and decision-making processes.

Therefore, this project proposes the development of a Smart Market and Price Decision Support System using Multi-Source Market Data Analytics and USSD Technology, with Ifakara serving as the case study area. The proposed system will collect market information from multiple digital and physical sources, integrate and analyze the collected data using machine learning techniques, predict future commodity prices, identify market opportunities, and provide intelligent recommendations to entrepreneurs, farmers, buyers, and other stakeholders. In addition, the system will provide market information through both web-based and USSD-based platforms to ensure accessibility among users regardless of internet availability. The implementation of the proposed system is expected to improve access to market intelligence, enhance decision-making capabilities, increase market efficiency, and contribute to sustainable socio-economic development within Ifakara and similar communities.

1.2. PROBLEM STATEMENT

Access to accurate, timely, and reliable market information is essential for effective decision-making among farmers, entrepreneurs, buyers, and other stakeholders involved in agricultural and business activities. In Ifakara and many other developing communities, market participants often depend on informal sources of information such as word-of-mouth communication, local traders, personal networks, and physical market visits to obtain information about commodity prices, market demand, supply conditions, and business opportunities. These methods are frequently unreliable, time-consuming, and incapable of providing real-time market intelligence.

The lack of an efficient market information system creates information asymmetry between market participants, where some individuals possess more market knowledge than others. As a result, farmers may sell their produce at lower prices due to limited awareness of prevailing market conditions, while buyers and entrepreneurs may struggle to identify competitive prices, profitable products, and emerging market opportunities. Consequently, stakeholders experience reduced profitability, poor bargaining power, inefficient resource allocation, and increased uncertainty in business decision-making (Galtier et al., 2014; Shepherd, n.d.).

Although various digital platforms and online marketplaces provide market-related information, most existing solutions focus on isolated sources of data and do not integrate information from multiple market channels. Furthermore, many systems lack advanced analytical capabilities such as market trend analysis, price forecasting, demand prediction, and intelligent recommendation generation. In addition, internet-based platforms are often inaccessible to rural users who rely on basic mobile phones and have limited internet connectivity (GSMA, 2023).

In Ifakara, there is currently no integrated platform that collects market information from multiple sources, analyzes the data using machine learning techniques, predicts future commodity prices, identifies business opportunities, and disseminates market intelligence through both web-based and USSD-based channels. The absence of such a system limits the ability of farmers, entrepreneurs, and buyers to make informed and data-driven decisions.

Therefore, there is a need to develop a Smart Market and Price Decision Support System that utilizes multi-source market data analytics, machine learning, and USSD technology to provide timely market information, price predictions, business recommendations, and decision support services. The

proposed system aims to improve market transparency, enhance accessibility to market intelligence, support evidence-based decision-making, and ultimately contribute to improved economic outcomes for farmers, entrepreneurs, and buyers in Ifakara.

1.3. PROJECT OBJECTIVE

1.3.1. GENERAL OBJECTIVE

The main objective of this study is to develop a Smart Market and Price Decision Support System that collects, analyzes, predicts, and disseminates market information from multiple data sources to support entrepreneurs, farmers, buyers, and other stakeholders in making informed and data-driven decisions.

1.3.2. SPECIFIC OBJECTIVE

To achieve the general objective, the project will focus on the following specific milestones:

- i. To develop a User Management Subsystem for user registration, authentication, and access control.
- ii. To develop a Market Data Collection Subsystem for gathering market information from multiple sources.
- iii. To develop a Data Scraping and Integration Subsystem for extracting and integrating market-related data from digital platforms.
- iv. To develop a Centralized Market Database Subsystem for storing, organizing, and managing market data.
- v. To develop a Machine Learning-Based Market Analysis Subsystem for identifying market trends, demand patterns, and business opportunities.
- vi. To develop an AI-Based Price Prediction and Recommendation Subsystem for forecasting commodity prices and generating recommendations.
- vii. To develop a USSD-Based Market Information Access Subsystem for providing market information through mobile phones without internet connectivity.
- viii. To develop a Reporting, Visualization, and Decision Support Subsystem for generating reports, dashboards, and analytical insights.

1.4. SCOPE OF THE STUDY

The scope of this project is defined by the following parameters to ensure the system is both effective and achievable within the given constraints:

- i. Focuses on entrepreneurs, farmers, and buyers, the system is designed to support market-related decision-making for these stakeholders.
- ii. Covers Ifakara as the case study area, where data collection, system implementation, and evaluation will be conducted.
- iii. Collects market information from multiple sources, including market officers, traders, online marketplaces, social media platforms, and publicly available data sources.
- iv. Integrates and stores market data in a centralized database to ensure data consistency, accessibility, and reliability.
- v. Uses machine learning techniques to analyze market trends, demand patterns, and business opportunities.
- vi. Provides commodity price prediction and recommendations to support production planning, pricing, purchasing, and investment decisions.
- vii. Uses USSD technology, enabling users to access market information through basic mobile phones without requiring internet connectivity.
- viii. Provides dashboards, reports, charts, and visualizations for market monitoring and decision support.

1.5. SIGNIFICANCE OF THE STUDY

The significance of this study can be explained as follows:

- i. Improves access to timely and reliable market information for entrepreneurs, farmers, and buyers, enabling them to make informed decisions.
- ii. Supports informed decision-making through market trend analysis, price predictions, and intelligent recommendations.
- iii. Enhances market transparency by reducing information asymmetry between farmers, traders, entrepreneurs, and buyers.
- iv. Enables users in rural areas to access market information through USSD technology without requiring internet connectivity.

- v. Assists farmers in planning production and marketing activities based on market demand and price forecasts.
- vi. Helps entrepreneurs identify business opportunities, market trends, and profitable investment areas.
- vii. Provides valuable market insights and analytical reports that can support planning and policy formulation by relevant authorities.
- viii. Promotes the adoption of modern digital technologies, artificial intelligence, and data analytics in agricultural and business sectors.

1.6. LIMITATION OF THE STUDY

The study is subject to the following limitations based on the proposed system design:

- i. The study is limited to Ifakara as the case study area.
- ii. The accuracy of the system depends on the availability and quality of market data collected from various sources.
- iii. Limited internet connectivity may affect data collection and updates from online platforms.
- iv. Price predictions may be affected by unforeseen factors such as weather conditions, economic changes, and market fluctuations.
- v. The study does not cover online payments, logistics management, or direct trading activities between buyers and sellers.

1.7. CHAPTER SUMMARY

This chapter introduced the study by providing background information on the role of Information and Communication Technology (ICT), Market Information Systems (MIS), machine learning, data analytics and USSD technology in supporting decision-making processes. The chapter highlighted the importance of market information in enhancing market transparency, reducing information asymmetry, and improving economic outcomes for entrepreneurs, farmers, and buyers. It also discussed the challenges associated with accessing timely, accurate, and reliable market information, particularly in rural areas such as Ifakara.

Furthermore, the chapter presented the problem statement, which justified the need for developing a Smart Market and Price Decision Support System capable of collecting, analyzing, predicting, and disseminating market information from multiple sources. The chapter also outlined the general objective and specific objectives that guide the development of the proposed system. In addition, the

scope, significance, and limitations of the study were presented to define the boundaries, expected benefits, and constraints of the project.

Overall, this chapter provides the foundation for the study and establishes the rationale for the development of the proposed system. The next chapter reviews related literature, theories, concepts, and existing systems relevant to market information systems, decision support systems, machine learning applications, and USSD-based information services, with the aim of identifying knowledge gaps and guiding the design of the proposed solution.

CHAPTER TWO

LITREATURE REVIEW

2.1. INTRODUCTION

This chapter presents a review of literature related to the development of a Smart Market and Price Decision Support System for entrepreneurs, farmers, and buyers. Literature review is an important part of research because it provides an understanding of existing knowledge, technologies, methodologies, and systems related to the study. It also helps identify gaps in previous studies and existing systems, thereby providing justification for the proposed solution.

The chapter examines various concepts related to Market Information Systems (MIS), Decision Support Systems (DSS), Machine Learning (ML), Data Analytics, Price Prediction, and USSD Technology. These concepts form the foundation upon which the proposed system is developed. The review also explores how modern technologies are being used to collect, process, analyze, and disseminate market information to support decision-making among stakeholders in agricultural and business sectors.

Furthermore, the chapter reviews existing market information systems and digital platforms that provide market-related services to farmers, traders, entrepreneurs, and buyers. The strengths, weaknesses, and limitations of these systems are analyzed to identify areas that require improvement. Particular attention is given to systems that provide market price information, forecasting services, mobile accessibility, and decision support functionalities.

In addition, the chapter reviews empirical studies conducted by various researchers on market information systems, market intelligence, commodity price prediction, and the application of machine learning in market analysis. The findings from these studies are used to understand current trends, challenges, and opportunities in the field of market information management and decision support.

Finally, the chapter identifies the research gap that exists in current systems and studies and presents a conceptual framework that illustrates the relationship between the major components of the proposed Smart Market and Price Decision Support System. The reviewed literature provides a theoretical and practical foundation that guides the design, development, and implementation of the proposed system.

2.2. EXISTING MARKET INFORMATION SYSTEMS

Market Information Systems (MIS) are computerized platforms designed to collect, process, store, analyze, and disseminate market-related information to support decision-making among farmers, traders, entrepreneurs, buyers, and policymakers. Over the years, several market information systems have been developed to improve market transparency, enhance access to market information, and reduce information asymmetry among market participants. This section reviews some existing market information systems relevant to the proposed study.

Table 2.1 presents a review of selected market information systems that have been implemented to support farmers, traders, buyers, and other stakeholders in accessing market-related information.

Table 1 Existing Market information system

System	Country	Strengths	Limitations
MFarm (Abass, 2013)	Kenya	Improves price transparency and bargaining power for farmers through mobile-based market information services.	Requires mobile application and internet access; limited accessibility for users with basic mobile phones.
eNAM (Pawar & Walke, 2024)	India	Enhances market efficiency, transparency, and competitive pricing by connecting multiple agricultural markets through a unified platform.	Internet-dependent and requires digital literacy, smartphones, or computer access.
Esoko Market Information System (David-Benz et al., 2014)	Ghana	Provides real-time market prices, weather information, agricultural advisory services, and mobile-based access to farmers.	Limited machine learning capabilities and lacks advanced price prediction and recommendation features.
M-Kilimo System (United Republic of Tanzania, 2022)	Tanzania	Provides agricultural information and advisory services through mobile technology and supports communication between farmers and agricultural officers.	Limited market analytics, no AI-based price prediction, and limited integration of multiple market data sources.
One2Two Market Platform (One2Two, 2023)	Tanzania	Facilitates access to commodity prices, market trends, and trading opportunities for agricultural stakeholders.	Limited predictive analytics, lacks machine learning-based forecasting, and has limited support for offline users.

From the reviewed systems, it is evident that existing market information systems have significantly improved access to market information and enhanced market transparency. However, most of these systems focus primarily on information dissemination and trading support, with limited capabilities in machine learning-based market analysis, price prediction, intelligent recommendation generation, and integration of multiple market data sources. Furthermore, some systems depend heavily on internet connectivity, limiting accessibility for rural users. Therefore, the proposed Smart Market and Price Decision Support System seeks to address these limitations by integrating multi-source market data analytics, machine learning-based price prediction, intelligent recommendations, and USSD technology to improve accessibility and decision-making among entrepreneurs, farmers and buyers.

2.2.1. LITREATURE REVIEW GAP IN EXISTING MARKET INFORMATION SYSTEMS

The review of existing market information systems such as M-Kilimo, Esoko, and eNAM shows that these systems have played an important role in improving access to market information for farmers, traders, buyers, and other stakeholders. They provide useful services such as market price updates, agricultural advisory information, weather forecasts, and market linkage opportunities. Through these services, users are able to access important information that helps them make better business and farming decisions.

However, despite their contributions, these systems still have several shortcomings. Most of them mainly focus on providing market information and advisory services without offering advanced features that support intelligent decision-making. For example, systems like M-Kilimo and Esoko provide users with market information but have limited capabilities for analyzing market trends, predicting future commodity prices, and generating recommendations that can help users make informed decisions.

Another challenge is that many existing systems collect information from only a few sources and do not fully combine data from different digital platforms such as social media, online marketplaces, e-commerce platforms, and public APIs. Because of this, users may not receive a complete picture of market conditions and emerging business opportunities. In addition, most of these systems provide limited reporting and visualization tools that can help users understand market trends and patterns more effectively.

Accessibility is also a concern. Some systems require internet access and a certain level of digital literacy, which can be challenging for farmers and rural stakeholders who use basic mobile phones and have limited internet connectivity. As a result, many potential users are unable to benefit fully from the available market information services.

Therefore, there is a need for a more advanced and integrated market information system that can collect data from multiple sources, analyze market trends using machine learning techniques, predict future commodity prices, and provide useful recommendations to users. The proposed Smart Market and Price Decision Support System aims to address these challenges by combining multi-source market data analytics, artificial intelligence, price prediction, reporting tools, and USSD technology into a single platform. This will enable entrepreneurs, farmers, and buyers in Ifakara to access reliable market information and make better decisions based on accurate and timely insights.

2.2.2. BLOCK DIAGRAM OF THE EXISTING SYSTEM (LEOLEO APP)

The M-Kilimo system is a mobile-based agricultural information platform designed to improve communication and information sharing among farmers, buyers, traders, extension officers, and other agricultural stakeholders. The system can be accessed through SMS, USSD, and web/mobile applications, making it available to both smartphone users and users with basic mobile phones.

The workflow begins when users such as farmers, buyers, and traders submit requests or access agricultural and market information through the available communication channels. The requests are processed by the M-Kilimo platform, which acts as the central component responsible for managing services and facilitating information exchange among stakeholders. The platform receives and integrates information from various sources, including extension officers, weather service providers, market information providers, and government agricultural institutions.

The collected information is stored in a centralized database where market data, weather information, advisory content, and user information are organized and managed. The platform then provides several services, including market information services, agricultural advisory services, market linkage

services, and communication services. Through these services, users can obtain market prices, agricultural advice, weather updates, and information about potential buyers and sellers.

Finally, the processed information and alerts are delivered back to users through SMS, USSD, or web/mobile applications, enabling them to make informed decisions regarding agricultural production, marketing, and business activities. This helps improve access to agricultural information, enhance market transparency, and strengthen communication between agricultural stakeholders.

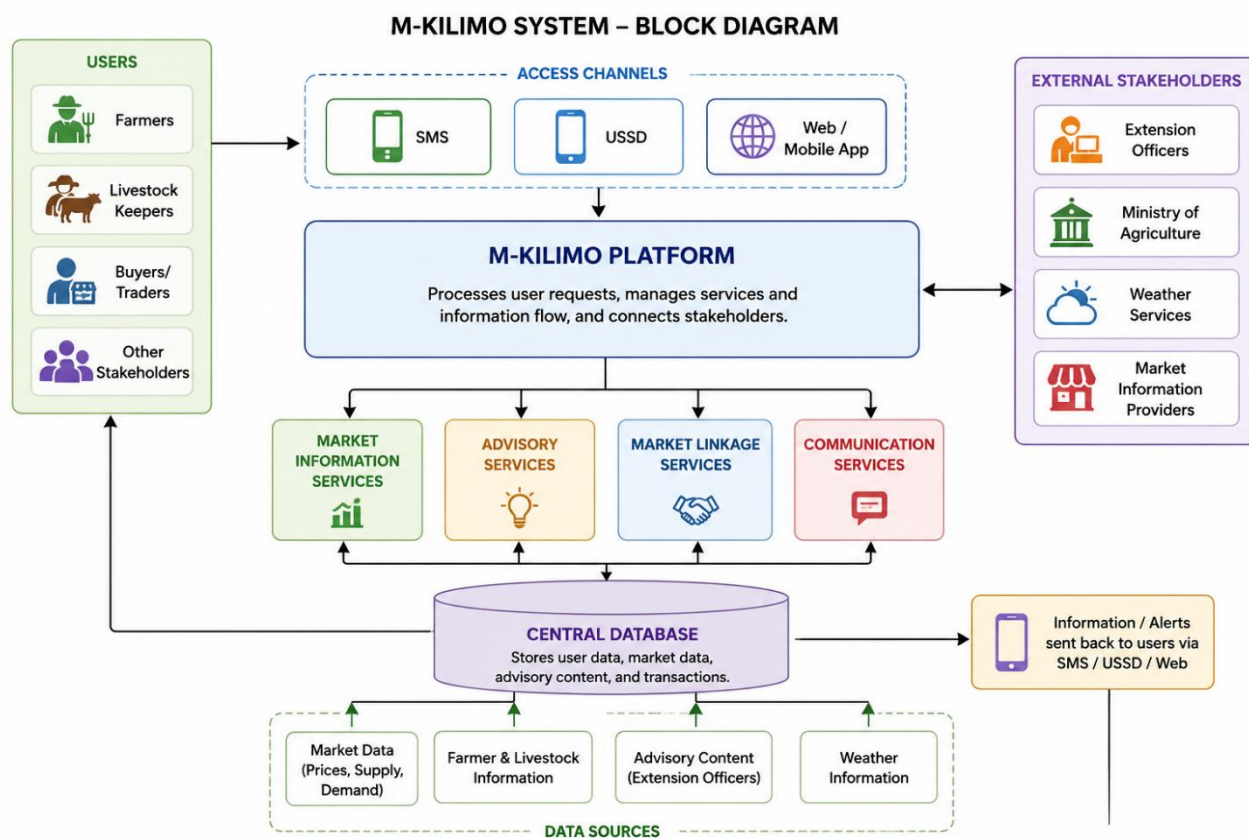


Figure 2.1 Block Diagram of M-Kilimo system

2.3. PROPOSED SYSTEM OVERVIEW

The proposed Smart Market and Price Decision Support System is a web-based and USSD-enabled platform designed to support entrepreneurs, farmers, and buyers by providing timely, accurate, and reliable market information. The system integrates market data from multiple sources, including market officers, traders, online marketplaces, social media platforms, e-commerce platforms, and publicly available APIs. By collecting and consolidating information from these sources, the system provides users with comprehensive market insights to support informed decision-making.

The system consists of several interconnected subsystems that work together to collect, process, analyze, store, and disseminate market information. The User Management Subsystem manages user registration, authentication, authorization, and role-based access control. The Market Data Collection Subsystem gathers market information from various sources, while the Data Scraping and Integration Subsystem automatically extracts and integrates relevant market data such as commodity prices, product demand indicators, customer reviews, and market trends.

The collected data is stored and managed within a Centralized Market Database Subsystem, which ensures data consistency, security, and reliability. The Machine Learning-Based Market Analysis Subsystem analyzes the collected data to identify market trends, demand patterns, and potential business opportunities. In addition, the AI-Based Price Prediction and Recommendation Subsystem forecasts future commodity prices and generates recommendations to assist users in making production, marketing, purchasing, and investment decisions.

To ensure accessibility for all users, including those in rural areas with limited internet connectivity, the system incorporates a USSD-Based Market Information Access Subsystem. This enables users to retrieve market prices, forecasts, recommendations, and market insights using basic mobile phones without requiring internet access. Furthermore, the Reporting, Visualization, and Decision Support Subsystem generates dashboards, charts, reports, and statistical summaries that provide meaningful insights into market conditions and support strategic decision-making.

Overall, the proposed system aims to improve market transparency, enhance access to market intelligence, reduce information asymmetry, and support data-driven decision-making among entrepreneurs, farmers, and buyers in Ifakara.

2.3.1. KEY COMPONENTS OF THE PROPOSED SYSTEM

The proposed Smart Market and Price Decision Support System consists of several key components that work together to collect, process, analyze, store, and disseminate market information. These components are designed to support entrepreneurs, farmers, and buyers in making informed and data-driven decisions.

i. User Management Component

This component is responsible for managing users of the system. It supports user registration, login, authentication, authorization, profile management, and role-based access control. Different user categories such as administrators, farmers, entrepreneurs, and buyers are assigned appropriate permissions based on their roles within the system.

ii. Market Data Collection Component

This component collects market information from multiple sources, including market officers, traders, online marketplaces, social media platforms, e-commerce platforms, and publicly available APIs. The collected information forms the foundation for market analysis and decision support.

iii. Data Scraping and Integration Component

This component automatically extracts market-related data from various digital platforms and integrates it into a unified format. The component gathers information such as commodity prices, market trends, customer reviews, product demand indicators, and other relevant market data.

iv. Centralized Market Database Component

This component stores and manages all collected market information within a centralized database. It ensures data consistency, security, accessibility, and reliability by organizing and maintaining market data for future retrieval and analysis.

v. Machine Learning-Based Market Analysis Component

This component analyzes collected market data using machine learning techniques. It identifies market trends, demand patterns, consumer preferences, seasonal variations, and potential business opportunities that can assist users in making informed decisions.

vi. AI-Based Price Prediction and Recommendation Component

This component uses historical and current market data to predict future commodity prices and generate recommendations. The generated forecasts and recommendations assist farmers, entrepreneurs, and buyers in planning production, marketing, purchasing, and investment activities.

vii. USSD-Based Market Information Access Component

This component enables users to access market information using mobile phones through USSD technology. It allows users, especially those in rural areas without internet access, to retrieve commodity prices, forecasts, recommendations, and market insights using basic mobile devices.

viii. Reporting, Visualization, and Decision Support Component

This component generates dashboards, reports, charts, graphs, and statistical summaries based on analyzed market data. The information presented helps users monitor market conditions, evaluate trends, and make strategic business decisions.

The integration of these components enables the proposed system to provide a comprehensive market intelligence platform capable of collecting data from multiple sources, performing advanced analysis, forecasting commodity prices, and delivering market information through both web-based and USSD-based channels. This approach enhances market transparency, improves accessibility, and supports effective decision-making among entrepreneurs, farmers, and buyers.

2.3.2. BLOCK DIAGRAM OF THE PROPOSED SYSTEM

The proposed Smart Market and Price Decision Support System is designed to collect, process, analyze, and disseminate market information to entrepreneurs, farmers, and buyers. The system gathers data from multiple sources, including market officers, traders, social media platforms, online marketplaces, e-commerce platforms, and public APIs. The collected data is stored in a centralized

database and analyzed using machine learning techniques to identify market trends, predict commodity prices and generate recommendations.

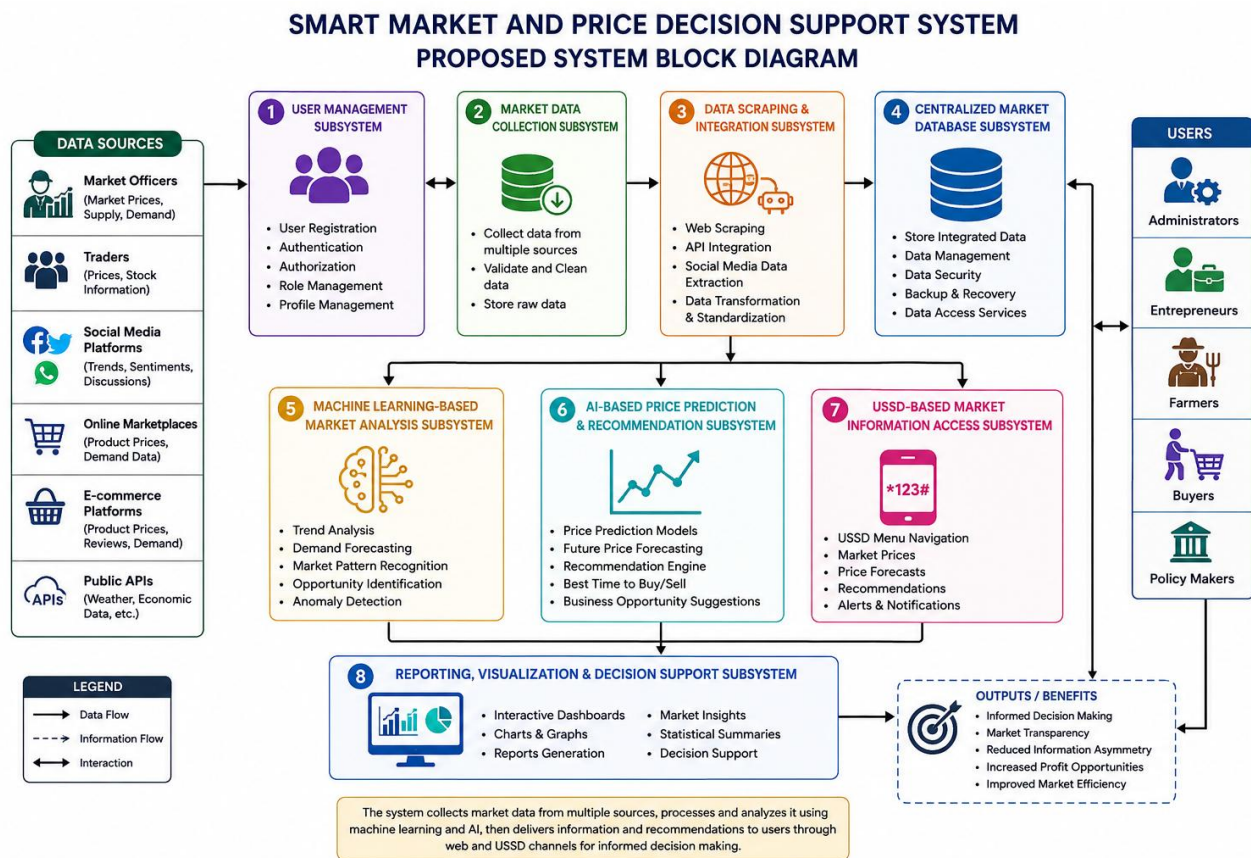


Figure 2.2 Block diagram of a proposed system

Users can access market information through both web-based and USSD-based platforms, ensuring accessibility for users with or without internet connectivity. The system also provides reports, dashboards, charts, and visualizations to support informed decision-making and market monitoring.

The proposed system addresses existing challenges by providing:

- i. Multi-source market data collection and integration.
- ii. Machine learning-based market analysis and price prediction.
- iii. Intelligent recommendations for decision-making.
- iv. Access to market information through both web and USSD technologies.
- v. Centralized data management for improved reliability and security.

CHAPTER THREE

METHODOLOGY

3.1. INTRODUCTION

This chapter presents the methodology used in the design, development, and implementation of the AI-Based Market Analysis and Decision Support System. It describes the research approach, system development methodology, data collection techniques, system design procedures, implementation strategies, and testing methods employed to achieve the objectives of the study.

The system is designed to provide comprehensive market intelligence by collecting, integrating, storing, analyzing, and disseminating market information to various stakeholders, including farmers, traders, consumers, researchers, and policymakers. The system incorporates multiple subsystems, namely the User Management Subsystem, Market Data Collection Subsystem, Data Scraping and Integration Subsystem, Centralized Market Database Subsystem, Machine Learning-Based Market Analysis Subsystem, AI-Based Price Prediction and Recommendation Subsystem, USSD-Based Market Information Access Subsystem, and Reporting, Visualization, and Decision Support Subsystem.

The proposed system utilizes modern technologies such as artificial intelligence, machine learning, web technologies, and USSD communication services to improve accessibility and support data-driven decision-making. Through predictive analytics and market trend analysis, the system aims to identify business opportunities, forecast commodity prices, and provide actionable recommendations to users.

This chapter further explains the system development approach, technologies used, and data collection techniques adopted to ensure that the developed system meets its functional and non-functional requirements effectively.

3.2. SYSTEM DEVELOPMENT

The methodology adopted for the development of the AI-Based Market Analysis and Decision Support System is the Agile Software Development Methodology.

Agile is an iterative and incremental software development approach that divides the project into small, manageable phases known as sprints. Each sprint focuses on developing and testing specific system functionalities, allowing continuous feedback from stakeholders and end users throughout the development process.

The selection of Agile methodology for this project is based on its flexibility and ability to accommodate changing requirements during system development. Since the project consists of several interconnected subsystems, including data collection, machine learning analysis, price prediction, USSD services, and reporting modules, Agile provides an effective framework for gradual implementation, integration, and evaluation of each subsystem.



Figure 3.1 Block diagram of a proposed system

The development of the AI-Based Market Analysis and Decision Support System will follow several structured phases to ensure systematic implementation and successful integration of all system components.

i. Requirements Gathering and Analysis

In this phase, structured questionnaires and interviews will be conducted with farmers, traders, consumers, and other stakeholders to collect information regarding current methods of accessing market information, challenges encountered in obtaining reliable market data, and their expectations

for the proposed system. The study will also identify issues related to price fluctuations, information asymmetry, and market inefficiencies. The collected data will be analyzed to determine user needs and to define the functional and non-functional requirements that will guide the development of the proposed system.

ii. System Planning and Technology Selection

During this phase, the overall system architecture will be planned to ensure accessibility, scalability, efficiency, and integration of artificial intelligence technologies. The major system modules will be identified, including the User Management Subsystem, Market Data Collection Subsystem, Data Scraping and Integration Subsystem, Centralized Market Database Subsystem, Machine Learning-Based Market Analysis Subsystem, AI-Based Price Prediction and Recommendation Subsystem, USSD-Based Market Information Access Subsystem, and Reporting and Visualization Subsystem. Appropriate technologies for backend development, database management, data integration, machine learning, and user interfaces will be selected to ensure compatibility and effective implementation of the proposed system.

iii. System Design

In this stage, a detailed technical blueprint of the system will be developed. System architecture diagrams, data flow diagrams, use case diagrams, and database models will be designed to illustrate the interactions among the various subsystems. The database schema will be designed to accommodate user information, market data, historical records, analytical results, and generated recommendations. The user interfaces, dashboards, and USSD menu structures will also be designed to ensure ease of use and efficient access to market information.

iv. System Development and Coding

During this phase, the various system modules will be implemented. The User Management Subsystem will be developed to support user registration, authentication, and access control. The Market Data Collection and Data Scraping Subsystems will be implemented to gather market information from multiple sources and integrate the collected data into the centralized database. Machine learning algorithms will be developed and trained using historical market data to identify trends, demand patterns, and business opportunities. Furthermore, artificial intelligence models for price prediction

and recommendation generation will be developed and integrated into the system. The web-based administrative portal, reporting tools, and USSD interface will also be implemented.

v. Testing, Deployment, and User Evaluation

In this phase, functional testing will be conducted to verify that all system components operate correctly and satisfy the specified requirements. Performance testing will be carried out to evaluate system reliability, responsiveness, and scalability. Usability testing will also be conducted with selected users to assess the ease of using the system and the usefulness of the information provided. After successful testing, the system will be deployed for pilot implementation, and user feedback will be collected for further improvements.

vi. Maintenance and Future Improvements

After deployment, the system will undergo continuous monitoring and maintenance to ensure reliable operation and to address any identified issues. User feedback and system performance reports will be used to introduce improvements and additional functionalities. Future enhancements may include the integration of additional market data sources, advanced predictive models, multilingual support, mobile applications, and enhanced decision-support capabilities to further improve the effectiveness of the system.

3.2.2. JUSTIFICATION FOR CHOOSING AGILE METHODOLOGY

The Agile methodology is particularly suitable for this project for the following reasons:

- i. **Flexibility** in Requirements: Farmer and trader needs may change during system development. This could include adding notifications, supporting multiple commodities, or improving analytics. Agile allows the system to respond to these changes efficiently.
- ii. **Iterative Development:** Features are developed incrementally. This approach ensures that functional modules, like USSD price submission and AI analysis are tested and refined continuously.
- iii. **User-Centered Feedback:** Frequent involvement of farmers, traders and administrators helps the system meet real user needs. This involvement ensures usability and accessibility.
- iv. **Early Testing and Deployment:** USSD interfaces and AI modules can be prototyped and tested early. This minimizes deployment risks and ensures reliability.

3.2.4. BENEFITS OF AGILE FOR THIS PROJECT

- i. **Adaptability:** The system can accommodate changes in user requirements or market conditions.
- ii. **Faster Delivery:** Core functionalities such as USSD price submission and query can be deployed early for immediate use.
- iii. **Higher User Satisfaction:** Continuous feedback ensures that the system meets the actual needs of farmers and traders.
- iv. **Reduced Risk:** Incremental development allows early detection of errors or usability issues before full deployment.

3.2.3 AGILE IMPLEMENTATION STEPS

The development of the AI-Based Market Analysis and Decision Support System is divided into five main sprints. Each sprint focuses on implementing and integrating specific system modules and functionalities.

Table 2 Features and Purpose of Each Sprint in Agile Methodology

S/N	FEATURES	PURPOSE
1	User Management Subsystem and System Foundation	Establishes user registration, authentication, authorization, and role-based access control while setting up the project's core architecture and database environment.
2	Market Data Collection, Data Scraping, and Centralized Database Subsystems	Enables the collection, extraction, integration, storage, and management of market information from multiple sources in a centralized database.
3	Machine Learning-Based Market Analysis and AI-Based Price Prediction Subsystems	Analyzes market trends, identifies demand patterns and business opportunities, predicts commodity prices, and generates intelligent recommendations for decision-making.
4	USSD-Based Market Information Access and Reporting Subsystems	Provides market information access through mobile phones without internet connectivity and generates reports, dashboards, visualizations, and analytical insights for users and administrators.
5	System Testing, Deployment, and User Evaluation	Verifies system functionality, performance, usability, and reliability, and incorporates user feedback to improve the system before final deployment.

3.3. DATA COLLECTION

The study will collect both primary and secondary data to support the development of the USSD-Based MIS. Primary data will be collected from farmers and traders through Structured Questionnaires to understand their challenges in accessing updated rice and beans prices and to identify system requirements. Secondary data will consist of historical rice and beans price records obtained from reliable market and agricultural sources. This data will be used to train and evaluate the selected machine learning models. Before analysis, the collected data will be cleaned, organized, and prepared to ensure accuracy and reliability. The combination of primary and secondary data will ensure that the system is both user-centered and technically effective.

Why using Questionnaire

- i. It allows collection of data from a large number of respondents efficiently.
- ii. It provides standardized responses suitable for machine learning processing.
- iii. It reduces interviewer bias.
- iv. It is cost-effective and time-efficient.
- v. It allows both quantitative and limited qualitative data collection.

The study will use purposive sampling to select farmers, traders and administrators who are directly involved in rice and beans production and trade in Morogoro. Respondents will be chosen based on their experience and relevance to the proposed USSD system. This approach will ensure that the collected data reflects real market challenges and user needs.

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